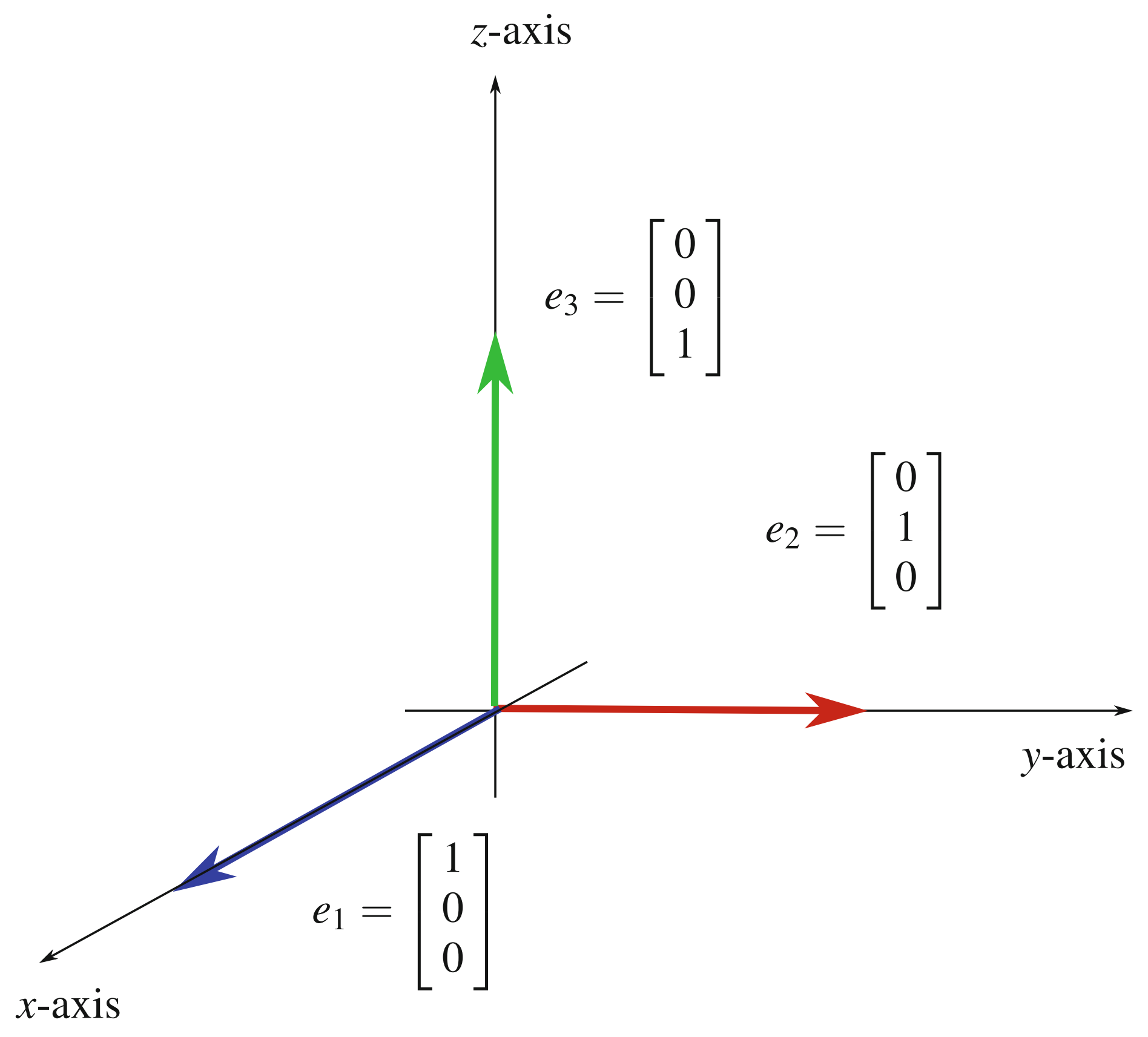

z-axis
$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
y-axis
$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
x-axis